

# Agamjot Nijjar

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## EDUCATION

UNIVERSITY OF TORONTO MISSISSAUGA | Mississauga, ON

2023 — 2028

Bachelor of Science, Geospatial Data Science

Coursework: Geographic Information Systems, Spatial Data Analysis, Computer Programming, Python for Data Science

## WORK EXPERIENCE

INVENTORY & DISTRIBUTION SYSTEMS ASSOCIATE — TJX Canada

May 2025 — August 2025

- Tasked with processing **200+** incoming and outgoing data center shipments per shift; maintained **99%** inventory accuracy by implementing a double-verification scanning workflow that eliminated mislabeling errors.
- Operated **industrial cutting machines** to process and cut tickets to specification, and performed hands-on **troubleshooting** of mechanical faults and calibration issues to minimize downtime and maintain uninterrupted production flow.
- Maintained detailed logs of hardware movements and flagged **12+** discrepancies over the term, enabling supervisors to resolve inventory conflicts before they escalated to audits.

COMPUTER SCIENCE INSTRUCTOR — Megamind Learning Centre

June 2024 — August 2024

- Delivered **daily lessons** in computer science and mathematics to groups of **15–20** students aged 8–14; developed visual, activity-based exercises that improved end-of-week assessment scores by **30%**.
- Taught a structured **troubleshooting** approach for common software and hardware issues, reducing instructor intervention requests by **25%** over the 8-week program.
- Communicated weekly progress updates to **parents and coordinators**, maintaining consistently positive feedback on student engagement and learning outcomes throughout the term.

## PROJECTS

PEEL REGION HOSPITAL COVERAGE MAPPER

Project 1

- Developed a **GIS-based population catchment analysis tool** in Python and ArcGIS Pro that mapped hospital service areas across Peel Region, Ontario, processing **1,746 dissemination area boundaries** for a population of **1.45 million** residents.
- Engineered a **coordinate system projection pipeline** using arcpy's PointGeometry and projectAs functions to convert hospital GPS coordinates from **WGS84** to **NAD83 Statistics Canada Lambert** projection, enabling accurate Euclidean distance calculations across all 1,746 spatial units.
- Processed and filtered a **740MB** Statistics Canada 2021 Census CSV using **pandas chunk-based reading** (100,000 rows per iteration) to extract Peel Region demographic data, joining population counts to dissemination area boundary shapefiles via **DAUID key matching**.
- Automated end-to-end **geodatabase population** using arcpy UpdateCursor, writing nearest-hospital assignments directly into ArcGIS feature class attributes and visualizing catchment zones as a **choropleth map** symbolized by unique hospital service areas in ArcGIS Pro.

ONTARIO 400 SERIES HIGHWAY TRAFFIC DENSITY MAP

Project 2

- Built an interactive **traffic density map** of Ontario's seven busiest 400-series highways in ArcGIS Pro and Python, splitting each mainline into individual stretches at every monitoring station rather than assigning one flat AADT value per route.
- Engineered a **spatial join pipeline** resolving MTO's internal **Linear Highway Referencing System (LHRS + Offset)** codes to real coordinates via the LHRS Base Points dataset, handling edge cases like combined route designations and non-standard highway codes.
- Classified stretches into **Low/Medium/High** volume tiers using fixed 2021 AADT tercile thresholds, then built a **1988–2021 historical time slider** in Leaflet.js and exported the final layers to **GeoJSON** as a single self-contained HTML/Leaflet.js page requiring no server, database, or GIS software.

## TECHNICAL SKILLS

**Programming & Data:** Python, C++ , Java, R, Folium, HTML, Data Visualization, Microcontroller Programming

**Tools & Software:** ArcGIS Pro, AutoCAD, Google Sketchup, GitHub, VS Code, Arduino, Circuit Design, R Studio

**Certifications:** Data Science 101 by IBM, Python 101 for Data Science by IBM, Intro to Machine Learning by Kaggle

**Languages:** English, Punjabi, Hindi